



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
M. Tech. Mechanical (Design Engineering)
Scheme 2024

S. Y. M. Tech. Mechanical (Design Engineering)
2026-27

Mechanical Engineering

Vision

- To produce world class globally competent distinguished graduates/ post graduates/ doctoral, Mechanical Engineers on the basis of their capabilities, dedication and work ethic and continuously strive towards societal development.

Mission

- **M1:** To impart quality Mechanical Engineering education in accordance with the needs of the society.
- **M2:** To produce globally competent Mechanical Engineers through research, industry institute interaction.
- **M3:** To help Mechanical Engineering graduates to implement their acquired engineering knowledge for society and community development.



Mechanical Engineering

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems
PO 2	An ability to write and present a substantial technical report/document
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the bachelor program of Mechanical Engineering.
PO 4	An ability to design mechanical systems by using principles of continuum mechanics, vibrations, reliability, maintainability, rapid prototyping & reverse engineering methods and feasibility of manufacturing processes.



PO 5	An ability to apply analytical and computational approach by employing software tools to design and develop a novel product and to optimize the existing ones.
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Department of Mechanical Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology



Mechanical Engineering

Course Code Format

M	D	U/P	2	C	C	1
Program Code		U-Under Graduate, P-Post Graduate	Semester No. / Year 1/2/3/4	Course Type		Course Serial No. 1-9

Program Code

MD	M. Tech Mechanical-Design Engineering
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Course Type

CC	Core Compulsory Course
EN*	Core Elective N* indicates the serial number of electives offered in the respective category
ON* Open Elective	N* indicates the serial number of electives offered in the respective category
SM	Seminar
DS	Dissertation
IN	Internship

Sample Course Code

MDP1CC1	Advanced Stress Analysis
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Mechanical-Design Engineering

M. Tech. Semester III

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
CMP3IK	Indian Science and Technology	2			2			50		50
MDP3IN	Internship/Field Project				3				100	100
MDP3DS1	Dissertation Phase –I			30	15			200	150	350
Grand Total		2		30	20			250	250	500

Mechanical-Design Engineering

M. Tech. Semester IV

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
CMP4OE1	Open Elective (SWAYAM / NPTEL MOOC) #				3					
MDP4DS2	Dissertation Phase –II			34	17		200		200	400
Grand Total				34	20		200		200	400



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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Second Year M.Tech. (Design Engineering), Semester-III

CMP3IK: Indian Science and Technology

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	----
Practical	----	ISE	50 Marks
Credits	2	ICA	----

Introduction:

This course will impart authentic knowledge of India's scientific and technological traditions. It would also try to provide an understanding of the socio-cultural and philosophical context in which the various scientific and technological ideas got developed in India and thereby help in repositioning India's contributions in science and technology.

Course Prerequisite:

Not required

Course Objectives:

The course intends to give a brief idea about the different aspects of ancient Indian technologies.

Course Outcomes:

After completing the course, students will be able to

1. Aware of the true history and rich culture of India
2. To explain the scientific value of the traditional knowledge of India
3. To appreciate the knowledge of Metallurgy, Textile Chemistry & Pyro Technology in IKS
4. To elaborate Indian knowledge of Water Management and transportation systems
5. To describe the contribution of IKS in Mathematics & Astronomy
6. To describe the concepts of IKS in Ecology and Environment.

Unit – I	Fundamentals	4 Hours
An overview of Indian contributions to technology. Technological Innovations.		
Unit – II	Metallurgy, Textile Chemistry & Pyro Technology	5 Hours
Copper/Bronze/Zinc: Important Mines (Zawar, Khetri mines), Iron and Wootz Steel Technology, Textile and Dyeing- Indian Specialities (Kutchi Embroidery, Cotton Textile etc.), Ceramic Technology, Stone (Lapidary), Shell, Ivory, Faience & Glass Technology		
Unit – III	Water Management & Transportation	5 Hours
Harappan and Traditional Water Management System of Gujarat, Historical Sites- Sringeverpur, South Indian Water Management System, Western Ghats Cave- Kanheri, etc., Communities Involved in Water Management, Modes of Transportations and Reforms, Grand Trunk		



Road (Uttarapath & Dakshinapath), Development of Trading Techniques, Boat & Ship Building		
Unit – IV	Mathematics & Astronomy	5 Hours
Mathematics contained in the Sulbasutra, Weaving Mathematics into Beautiful Poetry-Shankaracharya, The Evolution of Sine Function in India, The Discovery of Calculus by Kerala Astronomers, Vedanga Jyotish & Measuring Time & Calendar		
Unit – V	Ecology and Environment	5 Hours
Nakshatrara Gyaan and Agriculture, Vernacular Architecture, Forest Management and Urban Planning, Agroforestry, Tank, Lakes, and Stepwells		
Unit – VI	India's Contribution in Science and Technology to the World	4 Hours
Text Books		
Introduction To Indian Knowledge System: Concepts And Applications Mahadevan, B. · Bhat, Vinayak Rajat · Nagendra Pavana R.N. Apr 2022 · Phi Learning Pvt. Ltd.		
Reference Books		
1. R.M. Pujari, Pradeep Kolhe, N. R. Kumar, 'Pride of India: A Glimpse into India's Scientific Heritage', Samskrita Bharati Publication. 2. 'Indian Contribution to Science', compiled by Vijnana Bharati. 3. 'Knowledge traditions and practices of India', Kapil Kapoor, Michel Danino, CBSE, India. 4. Dr. Subhash Kak, Computation in Ancient India, Mount, Meru Publishing (2016) 5. Dharampal, Indian Science and Technology in the Eighteenth Century, Academy of Gandhian Studies, Hyderabad, 1971, republic. Other India Bookstore, Goa, 2000 6. Robert Kanigel, The Man Who Knew Infinity: A Life of the Genius Ramanujan, Abacus, London, 1999 7. Alok Kumar, Sciences of the Ancient Hindus: Unlocking Nature in the Pursuit of Salvation, CreateSpace Independent Publishing, 2014 8. B.V. Subbarayappa, Science in India: A Historical Perspective, Rupa, New Delhi, 2013 9. S. Balachandra Rao, Indian Mathematics and Astronomy: Some Landmarks, Jnana Deep Publications, Bangalore, 3rd edn, 2004 10. S. Balachandra Rao, Vedic Mathematics and Science in Vedas, Navakarnataka Publications, Bengaluru, 2019 11. Bibhuti bhushan Datta, Ancient Hindu Geometry: The Science of the Śulba, 1932, repr. Cosmo Publications, New Delhi, 1993 12. Bibhuti bhushan Datta & Avadhesh Narayan Singh, History of Hindu Mathematics, 1935, repr. Bharatiya Kala Prakashan, Delhi, 2004 13. George Gheverghese Joseph, The Crest of the Peacock, Penguin Books, London & New Delhi, 2000 14. J. McKim Malville & Lalit M. Gujral, Ancient Cities, Sacred Skies: Cosmic Geometries and City Planning in Ancient India, IGNCA & Aryan Books International, New Delhi, 2000). 15. Clemency Montelle, Chasing Shadows: Mathematics, Astronomy and the Early History of Eclipse Reckoning, Johns Hopkins University Press, 2011 16. Anisha Shekhar Mukherji, Jantar Mantar: Maharaja Sawai Jai Singh's Observatory in Delhi, 17. AMBI Knowledge Resources, New Delhi, 2010 18. Thanu Padmanabhan, (ed.), Astronomy in India: A Historical Perspective, Indian National		



Science Academy, New Delhi & Springer (India), 2010

19. Acharya Prafulla Chandra Ray, A History of Hindu Chemistry, 1902, republ., Shaibya Prakashan Bibhag, centenary edition, Kolkata, 2002
20. R. Balasubramaniam, Delhi Iron Pillar: New Insights, Indian Institute of Advance Study, Shimla & Aryan Books International, New Delhi, 2002
21. R. Balasubramanian, Marvels of Indian Iron through the Ages, Rupa& Infinity Foundation, New Delhi, 2008
22. Anil Agarwal & Sunita Narain, (eds), Dying Wisdom: Rise, Fall and Potential of India's Traditional Water-Harvesting Systems, Centre for Science and Environment, New Delhi, 1997
23. Fredrick W. Bunce: The Iconography of Water: Well and Tank Forms of the Indian Subcontinent, DK Print world, New Delhi, 2013

e-Resources

https://onlinecourses.swayam2.ac.in/arp19_ap87/preview





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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Second Year M.Tech. (Design Engineering), Semester-III

MDP3IN: Internship/Field Project

Teaching Scheme

Credits

3

Examination Scheme

ICA

100 Marks

Course Objectives:

1. To make students aware about the nature of problems existing in industries or society
2. To introduce students to various methodologies for solving the identified problems

Course Outcomes:

At the end of the course, students will be able to

1. Identify real-world problems related to Design Engineering
2. Solve the identified problem using appropriate methodology

Internship

Students are expected to undergo a rigorous training for a minimum period of one month during summer term/vacation at leading Engineering organizations/Research laboratories/Design and Consultancy organizations.

OR

Field Project

- Students should finalize a particular habitation or village or municipal ward, as far as possible, in the near vicinity of their place of stay.
- Students may work in close association with Non-Governmental Organizations like Lions Club, Rotary Club, etc. or with any NGO actively working in that habitation.
- Then, he/she should conduct a preliminary survey including the socio-economic conditions of the allotted habitation, in terms of their own domain or subject area. Or it can even be a general survey, incorporating all the different areas.
- If required, a survey form based on the type of habitation (rural, urban etc.) should be prepared before visiting the habitation.
- The Governmental agencies, like revenue administration, corporation and municipal authorities and village secretariats may be aligned for the survey.
- Analysis of the collected data should be done.
- A solution should be proposed to the problem identified.
- Student should prepare a report which should include following points.
- Students may take help from different government departments like Agriculture, Health, Marketing and Cooperation, Animal Husbandry, Irrigation etc.

Assessment

- The students are expected to present the work done during internship/ field project.
- The students shall submit a detailed report based on activities done in internship / field project and learnings through the same.
- The students shall also submit the duly signed certificate from the organization/s where internship/ field project was done, clearly indicating the period of internship/ field project in the certificate.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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Second Year M.Tech. (Design Engineering), Semester-III

MDP3DS1: Dissertation Phase –I

		Examination Scheme	
Engagement	30 Hours/week	ICA	150 Marks
Credits	15	ISE	200 Marks

Deliverables:

1. Recognize the importance of planning and preparation required to undertake a research project
 2. Search the existing literature and identification of research problem
 3. Design and develop the solution for complex engineering problem
 4. Demonstrate the ability to collate and critically assess/interpret data
 5. Develop an ability to effectively communicate knowledge in a scientific manner
1. Students are expected to carry out independent research work on the chosen topic. I
 2. In this semester it is expected that the student has carried out substantial research work including exhaustive literature survey, formulation of the research problem, based on which he/she has to submit the synopsis.
 3. Students have to give progress report at the end of semester and also communicate the progress of work during the semester at regular intervals to the guide.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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MDP4DS2: Dissertation Phase –II

		Examination Scheme	
Engagement	34 Hours/week	ICA	200 Marks
Credits	17	OE	200 Marks

Deliverables:

1. Recognize the importance of planning and preparation required to undertake a research project
2. Extend the research work to a standardized level that can give solutions to real life problems.
3. Demonstrate the ability to collate and critically assess/interpret data
4. Develop an ability to effectively communicate knowledge in a scientific manner

1. Students are expected to carry out independent research work on the chosen topic. In this semester it is expected that the student has carried out substantial research work including exhaustive literature survey, development/fabrication of experimental set-up (if any/required) and testing, and analysis of initial results thus obtained.
2. It is expected that the student has completed most of the experimental/computation works and analysed the results so obtained as proposed in the synopsis.
3. The work should be completed in all respects in this semester. The students are required to submit the dissertation work in the form of report as per the institute rule.
4. The report shall be hard bound. It is a report on the work done by the student. It should have literature review, problem definition and formulation, adopted methodology, experimentation plan if any, results and discussion, conclusions, and its relevance to the further work.
5. The viva voce examination of the Dissertation-II shall consist of a presentation by the candidate and demonstration of the work carried out.

